



CS614-FALL
ASSIGNMENT-1
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CS619 & CS519 PROJECTS

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QUESTION NO 1:

OLTP (OnLine Transaction Processing) specific query

Select tx_date, balance from tx_table

Where account_ID = 592156;

DWH specific query

Select balance, age, sal, gender from customer_table and tx_table

Where age between (20 and 30) and Education = 'graduate' and CustID.customer_table = Customer_ID.tx_table;

Let's take a brief look at these two queries, write the results by comparing them, and summarize them in the table below:

SOLUTION:

Criteria	OLTP Query	DWH Query
Primary Key Usage	Utilizes the primary key (account_ID) for quick access to transaction data.	Joins tables based on primary and foreign keys (CustID.customer_table = Customer_ID.tx_table) for filtering.
Number of Tables	Queries a single table (tx_table) for transaction data.	Queries multiple tables (customer_table and tx_table) to gather related information.
Rows Returned	Typically returns a small number of rows (usually a single row or few rows for specific transactions).	Returns a larger number of rows due to filtering conditions across joined tables (potentially thousands of rows based on criteria).
Query Selectivity	High selectivity since it targets a specific account_ID and retrieves precise data.	Lower selectivity due to broader conditions (age range, education) that may return many records.

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Indexing	Likely relies on indexing for fast lookups based on account_ID for quick transaction retrieval	May use indexes on multiple columns (age, education, Customer_ID) to optimize join and filtering performance, but may not be as efficient as OLTP indexing due to complexity.
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QUESTION NO 2:

Consider an employee database system developed for a multi-branch company. Each branch is based on a specific department in a different city, e.g., HR, Admin, or Head Office.

Table 1: Employee Record

EmployeeID	EmployeeName	Branch	Department	JobTitle	DateOfJoining	Salary
Emp32055	Salman Ahmed	Islamabad	Head office	CEO	2016-11-01	850000
Emp32001	Hussain Ali	Lahore	Amin	Admin Officer	2019-05-01	350000
Emp32056	Uzma khan	Lahore	Admin	Supervisor	2019-05-01	180000
Emp32022	Zafar Ali	Lahore	Admin	Admin Officer	2020-03-15	220000
Emp32035	Usama Khan	Karachi	HR	HR Assistant	2021-08-20	150000
Emp32077	Ali Ahmed	Karachi	HR	HR Manager	2021-05-01	280000

Identify the normalization form(NF) of the given table and write down the anomaly type of given statements.

SOLUTION:

Normalization Form(NF)	1NF
If a new branch is added (e.g., a new branch in another city), it is impossible to record it in the table unless an employee is also added.	Insertion Anomaly (can't add a branch without an employee)
If all employees from a particular branch are removed, information about that branch (e.g., the city and department) will also be lost.	Deletion Anomaly (losing branch info when all employees are deleted)
If a department or branch location of a specific department changes (e.g., the HR department moves to a different city), the change must be made in multiple rows. If any change is missed, it can lead to inconsistent data.	Update Anomaly (risk of inconsistent data if updates are missed)

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EXPLAIN:

Analysis of Normalization Forms

1. First Normal Form (1NF):

- **Definition:** A table is in 1NF if all columns contain atomic values (i.e., no repeating groups or arrays).
- **Evaluation:** The table meets this criterion since all attributes contain atomic values and there are no repeating groups.
- **Conclusion:** The table is in **1NF**.

2. Second Normal Form (2NF):

- **Definition:** A table is in 2NF if it is in 1NF and all non-key attributes are fully functionally dependent on the primary key. There should be no partial dependency of any column on a composite key.
- **Evaluation:** Assuming EmployeeID is the primary key, all attributes depend on it. However, the Branch and Department attributes could create partial dependencies since they might only depend on the job title or other factors (e.g., branches may have multiple job titles).
- **Conclusion:** The table is **not in 2NF** due to potential partial dependencies.

3. Third Normal Form (3NF):

- **Definition:** A table is in 3NF if it is in 2NF and all the attributes are functionally dependent only on the primary key, with no transitive dependencies.
- **Evaluation:** Since the table is not in 2NF, it cannot be in 3NF. Moreover, if Branch determines Department, it creates a transitive dependency.
- **Conclusion:** The table is **not in 3NF**.

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